Cross-Section Comparison: Latest vs Existing using RS67

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Input ROOT Files Used

Data Files:

- `merged_RS67_3089_LH2.root`
- `merged_RS67_3089_LD2.root`
- `merged_RS67_3089_Flask.root`

MC Files (Acceptance Correction):

- `rs67_lh2_acc.root`
- `rs67_ld2_acc.root`

MC Files (Reconstruction Efficiency):

- `rs67_lh2_eff.root`
- `rs67_ld2_eff.root`

Event Selection (chuck_cuts_2111-v42)

Dimuon Kinematics & Quality:

- $4.2 < M_{\mu\mu} < 8.8$ GeV (or dynamically adjusted)
- $-0.1 < x_F < 0.95$ (Standard integration window $0.0 \leq x_F < 0.80$)
- $0.05 < x_T \leq 0.58$, $|\cos\theta| < 0.5$
- Target-dump vertex separation: $z_{vtx} \in (-280, -5)$ cm
- Transverse momentum bounds & $\chi^2_{\mu\mu} < 18$

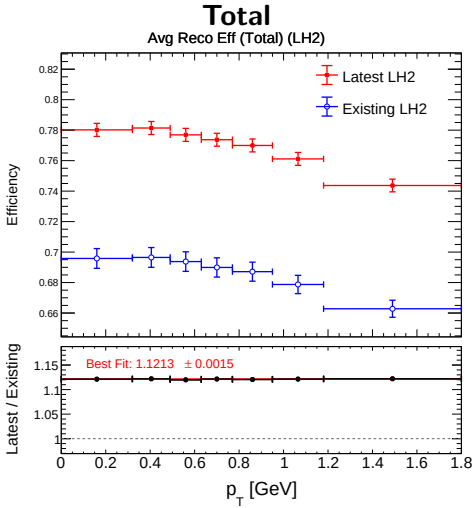
Single Track Quality:

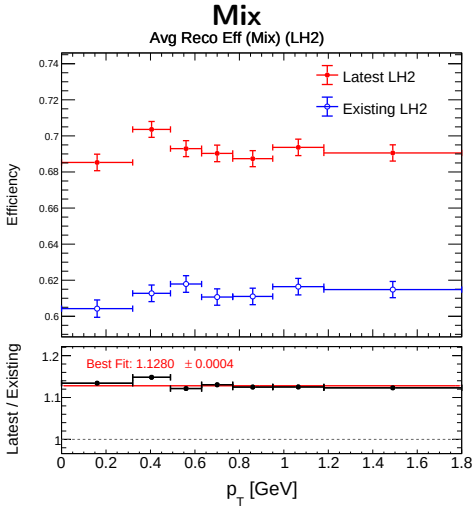
- $9 < p_{z,st1} < 75$ GeV, Total Hits > 13
- $\chi^2_{target} < 15$, Target & Dump spatial back-tracking matching
- p_T kick match across FMAG (≈ 0.416 GeV)

Combinatorial & Occupancy:

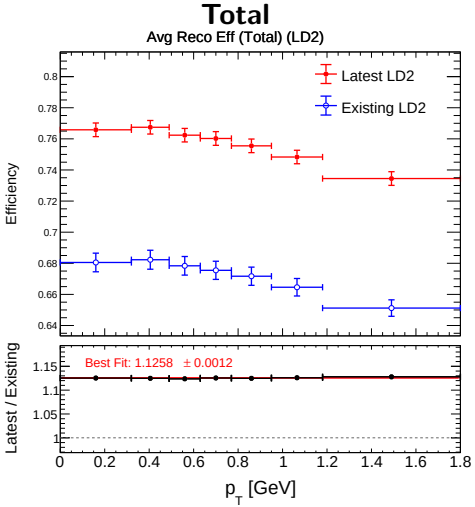
- Opposite half tracking ($y_{1,st3} \times y_{2,st3} < 0$)
- St1 total hits > 8, Overall total hits > 29
- Chamber Multiplicity: $D_1, D_2, D_3 < 400$ and $20 < D_{1'} < 385$

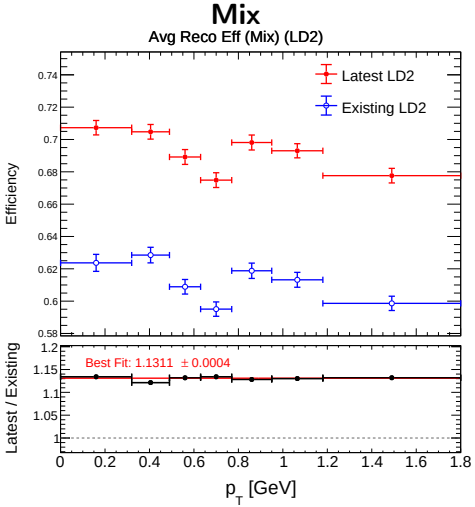
Reconstruction Efficiency Comparison (LH2)



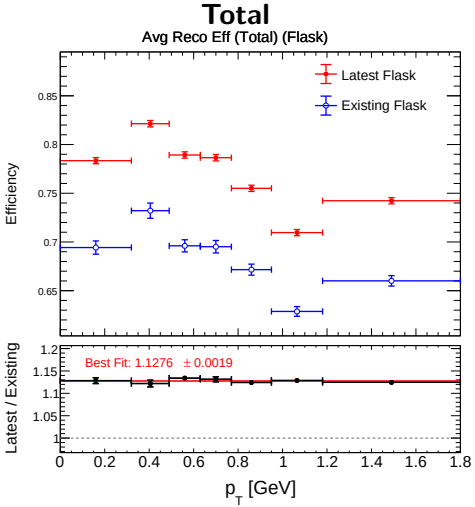


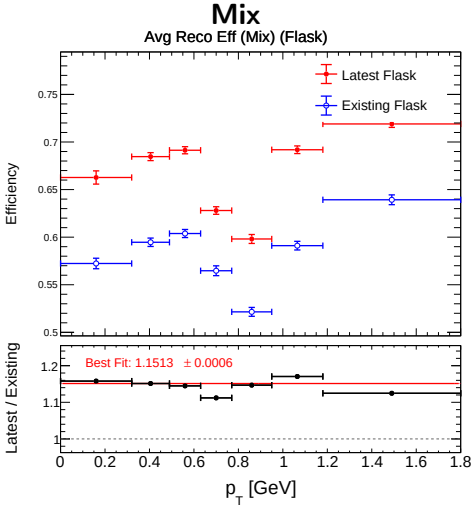
Reconstruction Efficiency Comparison (LD2)



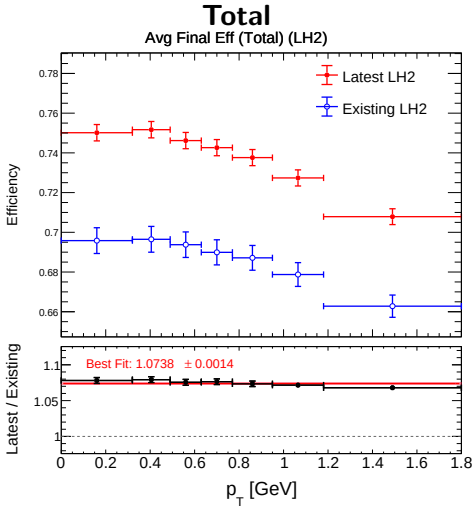


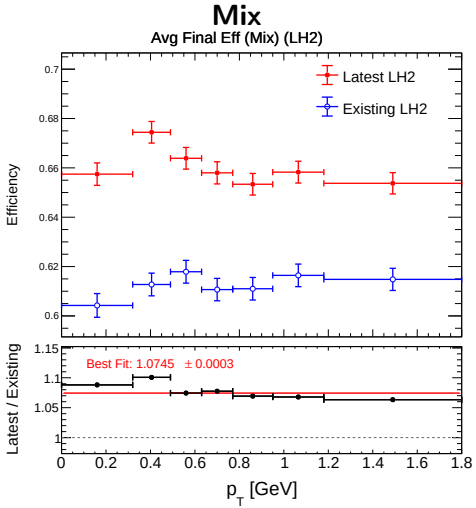
Reconstruction Efficiency Comparison (Flask)



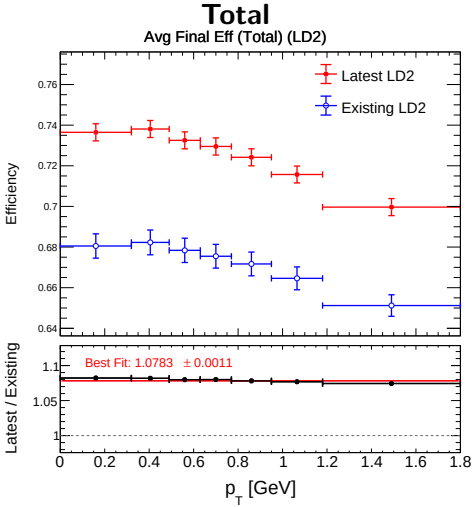


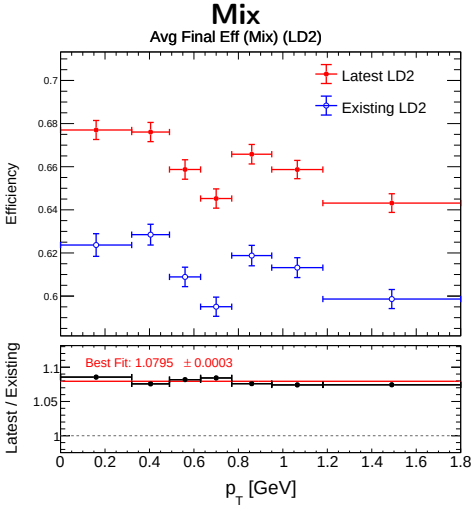
Final Efficiency Comparison (LH2)



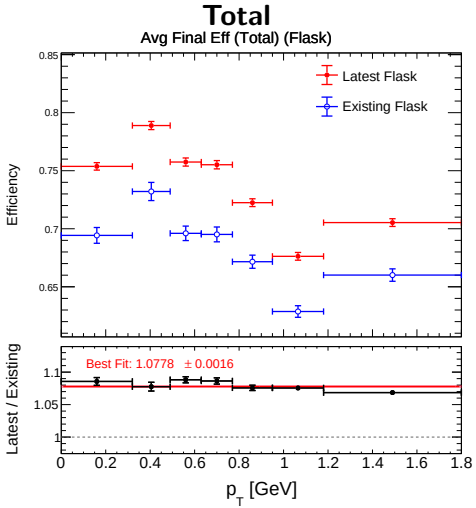


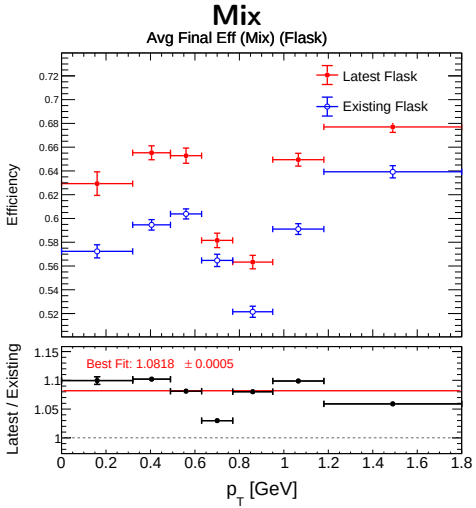
Final Efficiency Comparison (LD2)



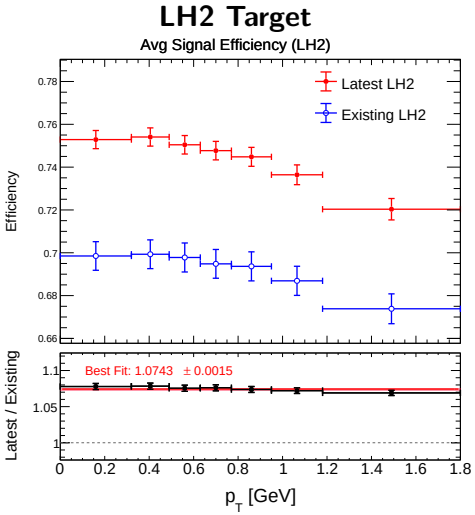


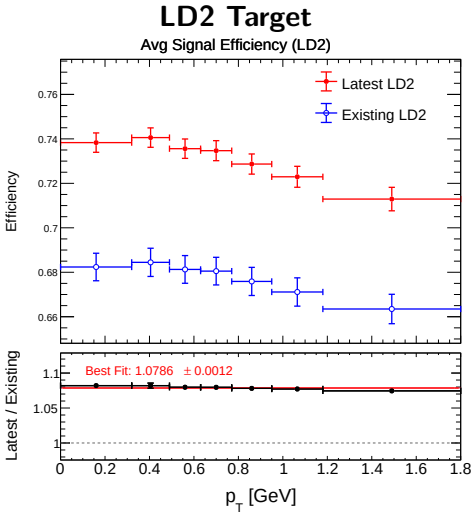
Final Efficiency Comparison (Flask)



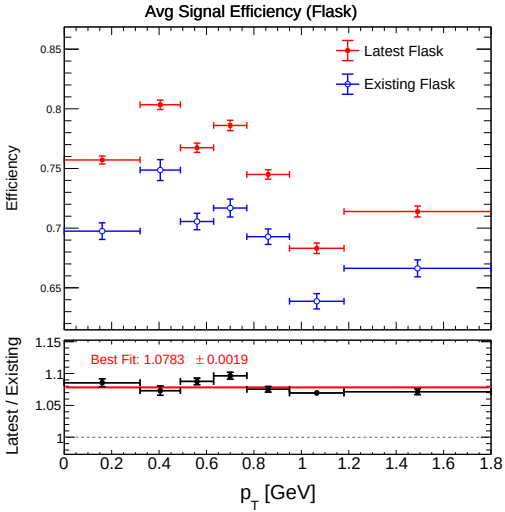


Signal Efficiency Comparison (LH2 vs LD2)



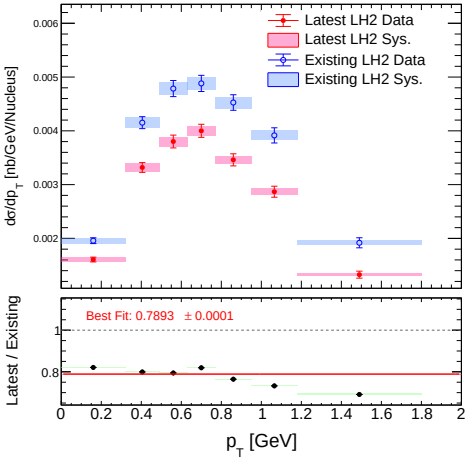


Signal Efficiency Comparison (Flask)



Single Differential Cross-Section Comparison

LH2 Target



LD2 Target

